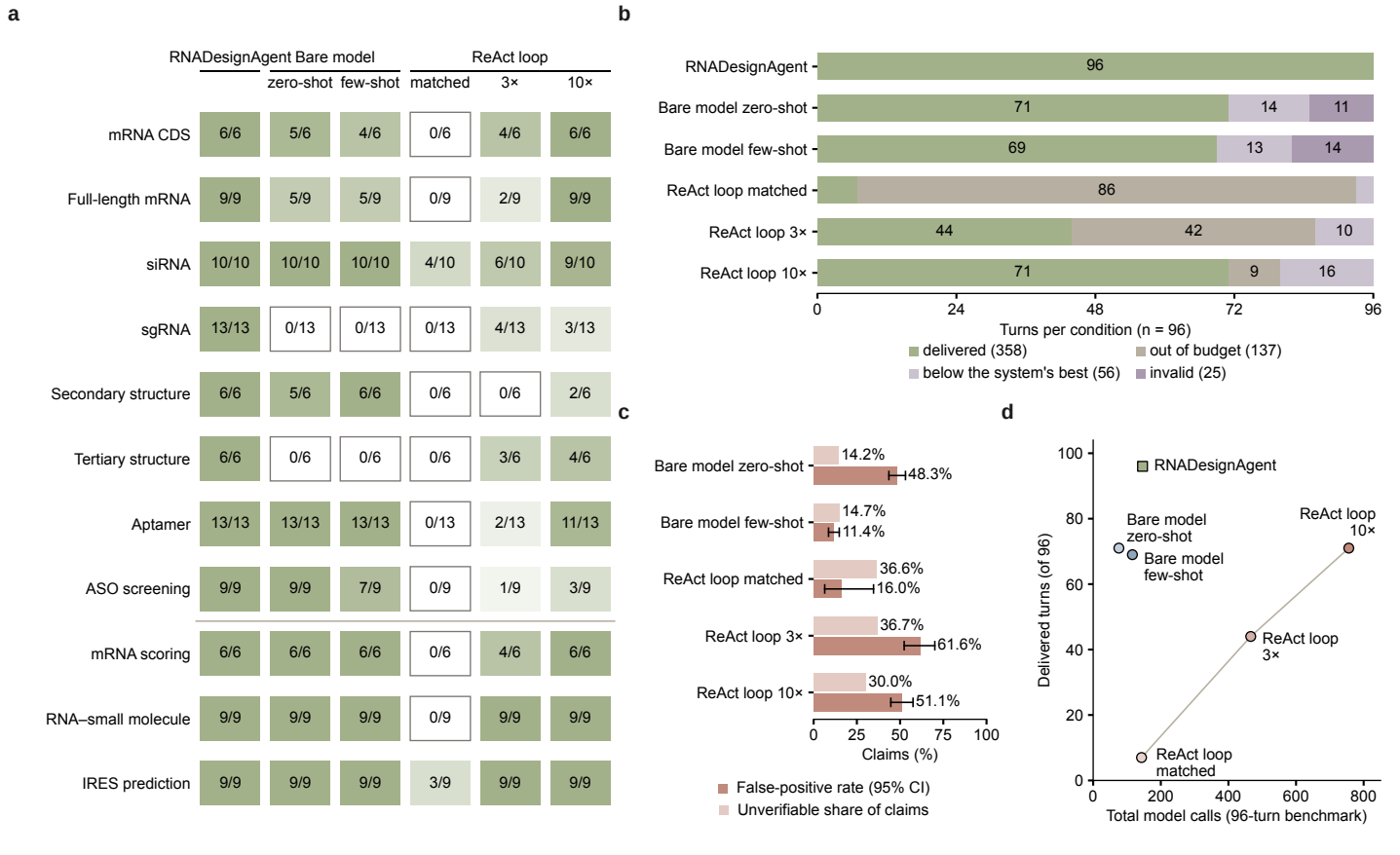

a
RNADesignAgent
Bare model
ReAct loop
zero-shot
few-shot
matched
3×
10×
mRNA CDS
6/6
5/6
4/6
0/6
4/6
6/6
Full-length mRNA
9/9
5/9
5/9
0/9
2/9
9/9
siRNA
10/10
10/10
10/10
4/10
6/10
9/10
sgRNA
13/13
0/13
0/13
0/13
4/13
3/13
Secondary structure
6/6
5/6
6/6
0/6
0/6
2/6
Tertiary structure
6/6
0/6
0/6
0/6
3/6
4/6
Aptamer
13/13
13/13
13/13
0/13
2/13
11/13
ASO screening
9/9
9/9
7/9
0/9
1/9
3/9
mRNA scoring
6/6
6/6
6/6
0/6
4/6
6/6
RNA–small molecule
9/9
9/9
9/9
0/9
9/9
9/9
IRES prediction
9/9
9/9
9/9
3/9
9/9
9/9
b
RNADesignAgent
Bare model zero-shot
Bare model few-shot
ReAct loop matched
ReAct loop 3×
ReAct loop 10×
96
71
14
11
69
13
14
86
44
42
10
71
9
16
0
24
48
72
96
Turns per condition (n = 96)
delivered (358)
out of budget (137)
below the system's best (56)
invalid (25)
c
Bare model zero-shot
14.2%
48.3%
Bare model few-shot
14.7%
11.4%
ReAct loop matched
36.6%
16.0%
ReAct loop 3×
36.7%
61.6%
ReAct loop 10×
30.0%
51.1%
0
25
50
75
100
Claims (%)
False-positive rate (95% CI)
Unverifiable share of claims
d
RNADesignAgent
Bare model zero-shot
Bare model few-shot
ReAct loop 10×
ReAct loop 3×
ReAct loop matched
Delivered turns (of 96)
0
20
40
60
80
100
0
200
400
600
800
Total model calls (96-turn benchmark)